



“ IT IS **FUTILE** TO DO WITH **MORE THINGS** THAT WHICH CAN BE DONE WITH **FEWER**. ”

William of Ockham, Franciscan friar, from *Summa Totius Logicae*, c.1323

“ GOD HIMSELF WAS A **PRACTICING SURGEON** WHEN HE **MADE** THE FIRST **MAN** FROM MUD AND **EVE** FROM ONE OF HIS RIBS. ”

Henri de Mondeville, French surgeon, from *Cyrurgia*, c.1312

produced his *Cyrurgia* (Surgery), a manual based in part on his observations of dissected corpses, although his definitions were not always accurate.

The **practice of dissection** was revived by Italian medical professor **Mondino de Luzzi** (c.1275–1326), who taught at Bologna, and who performed a **public dissection** in 1315.

Dissections featured regularly in his teaching. In 1316, Mondino completed *Anatomia*, the first textbook specifically concerned with anatomy (rather than surgery).

An atlas of the body

This anatomical drawing from Henri de Mondeville's *Cyrurgia* shows the lower part of the torso cut away, revealing the internal organs.



IN 1323, WILLIAM OF OCKHAM

produced one of the greatest works of logic of the Middle Ages—*Summa Logicae* (The Logic Handbook)—in which he radically diverged from traditional Christian philosophy. Most notable in William's ideas is the idea of economy, that if a cause or factor is unnecessary to prove an argument, then it should be discarded, a principle that came to be called **Ockham's razor**. He promoted the idea that individual perception is the foundation of all knowledge about the world, attacking long-held metaphysical explanations for the order of the Universe. He also advocated the separation of secular and ecclesiastical power.

Windmills had been used to grind flour since the 12th century in Europe, but in 1345 **windmills** are first recorded as being used to operate water-pumps to **drain land in the Netherlands**. The resulting reclaimed land, or polders, ultimately came to make up a fifth of the country, which is still protected from the sea by a system of dikes.

Around 1349, the French scholar **Nicholas of Oresme** (c.1320–82) expounded a **system using graphs to represent the growth of a function** (such as the velocity of an object), which was a great aid to mathematical



WILLIAM OF OCKHAM (c.1285–1349)

Franciscan friar William of Ockham studied at Oxford and by 1315 he was lecturing on the Bible. His theories of logic were seen by many as attacking Christian tenets, and he was summoned to the Papal court at Avignon to answer charges of erroneous teaching. He fled before the enquiry was concluded and spent the rest of his life at the court of the Holy Roman Emperor Louis IV of Bavaria.

analysis. Later, in 1377, he proposed the idea in his *Livre du Ciel et du Monde* (Treatise of the Sky and the World) that the Earth was not immobile at the center of the Universe, as traditional cosmology held, but that it **rotated on its axis**. He met objections from those who said birds would simply fly off it, by affirming that the oceans were included in the rotation.

20 PERCENT
THE PROPORTION OF THE **NETHERLANDS** THAT IS **RECLAIMED LAND**

1315 Mondino de Luzzi performs the first public dissection in Bologna

1316 De Luzzi publishes *Anatomia*, the first public textbook specifically on anatomy

1319 First cannon appears in Europe

1323 William of Ockham's *Summa Logicae* describes Ockham's razor

1343 First Public Health Commission established in Venice

c.1340 Englishman John of Dumbleton suggests that no parts of a substance are eliminated during condensation, paving way for a theory of molecules

1345 Windmills used for reclaiming land through drainage in the Netherlands

1348 English mathematician Richard Swineshead proves equations concerning movement of bodies with uniform velocity

c.1349 Nicholas of Oresme expounds a system of graphical representation